

19.5/20

MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY
COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY (COICT)
ET 8213 TEST ONE 2023/2024.

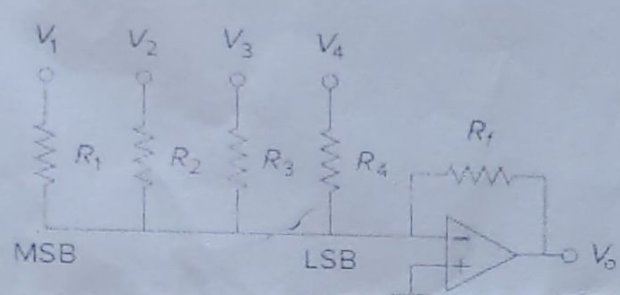
Reg no. 22100934340091 Time: 60min

QUESTION ONE [5 Marks]

- a) What is an Operational amplifier, what is the main purpose of negative feedback?
- b) Give four characteristics of ideal Opamp circuit.
- c) Draw and derive an equation for closed loop gain for non-inverting amplifier circuit configuration.

QUESTION TWO [5 Marks]

In the DAC circuit below, $R_f = 10\text{ k}\Omega$, $R_1 = 10\text{ k}\Omega$, $R_2 = 20\text{ k}\Omega$, $R_3 = 40\text{ k}\Omega$, and $R_4 = 80\text{ k}\Omega$. Obtain the analog output voltage for the digital inputs [0000], [1000], [0010], [0111], [1110]. Consider $V_{ref} = 1\text{ V}$.

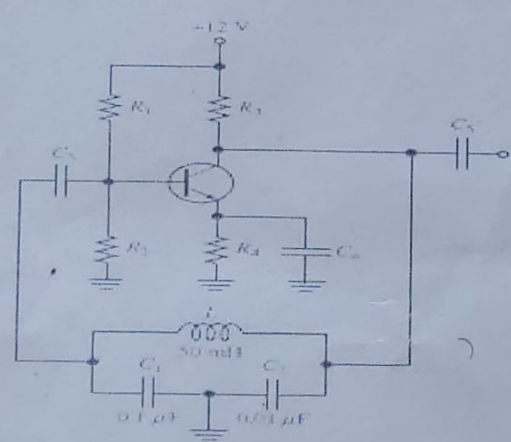


QUESTION THREE [5 Marks]

- a) What are the conditions required for a circuit to oscillate?
- b) Under what principle does square wave oscillator operate? Draw and briefly explain the circuit of square wave oscillator.

QUESTION FOUR [5 Marks]

- a) What is the basic difference between the Colpitts and the Hartley oscillators?
- b) Determine the frequency for the oscillator below assume there is negligible loading on the feedback circuit.

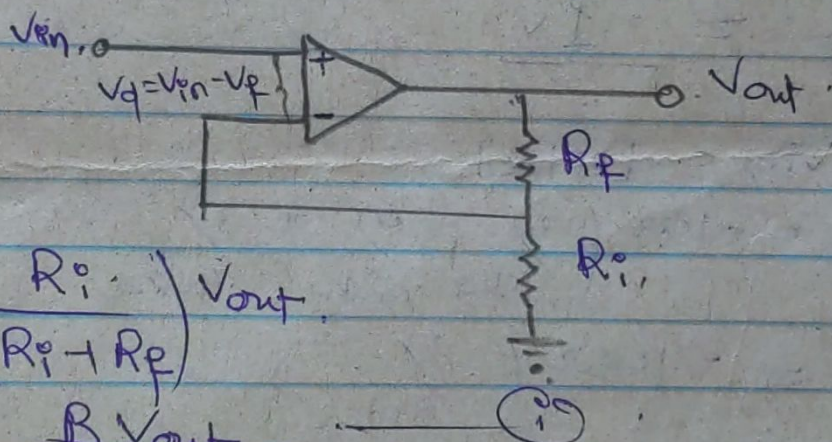


(a) Operational amplifier is the device which is used to perform mathematical operation such as addition, subtraction, integration and differentiation.

Purpose of negative feedback is to avoid infinite voltage gain in a circuit.

- (b). (i) It has infinite voltage gain.
 (ii) It has infinite bandwidth.
 (iii) It has infinite input impedance.
 (iv) It has zero output impedance.

(c) Consider,



$$V_d = \left(\frac{R_i}{R_i + R_f} \right) V_{out}$$

$$V_f = B V_{out}$$

$$V_{out} = A_d V_d$$

$$V_{out} = A_d (V_{in} - V_f)$$

$$V_{out} = A_d (V_{in} - B V_{out})$$

$$V_{out} = A_d V_{in} - A_d B V_{out}$$

$$V_{out} + A_d B V_{out} = A_d V_{in}$$

$$V_{out} (1 + A_d B) = A_d V_{in}$$

$$\frac{V_{out}}{V_{in}} = \frac{A_d}{1 + A_d B}$$

But $A_d B \gg 1$ then, $1 + A_d B = A_d B$

$$\frac{V_{out}}{V_{in}} = \frac{A_d}{A_d B} = \frac{1}{B} = \frac{R_i + R_f}{R_i}$$

$$A_d = 1 + \frac{R_f}{R_i}$$

2. Inverting amplifier.

$$V_{out} = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} + \frac{V_4}{R_4} \right) \quad (02)$$

for. first $V_{in} = [0000]$

$$V_{out} = -10k\Omega \left(\frac{0}{10k\Omega} + \frac{0}{20k\Omega} + \frac{0}{40k\Omega} + \frac{0}{80k\Omega} \right)$$

$$V_{out} = \underline{0V} \quad \checkmark \quad 0.5$$

for. $V_{in} = [1000]$

$$V_{out} = -10k\Omega \left(\frac{1V}{10k\Omega} + \frac{0}{20k\Omega} + \frac{0}{40k\Omega} + \frac{0}{80k\Omega} \right)$$

$$V_{out} = -10k\Omega \left(\frac{1V}{10k\Omega} \right)$$

$$V_{out} = \underline{-1V} \quad \checkmark \quad 0.5$$

for. $V_{in} = [0010]$

$$V_{out} = -10k\Omega \left(\frac{0}{10k\Omega} + \frac{0}{20k\Omega} + \frac{1V}{40k\Omega} + \frac{0}{80k\Omega} \right)$$

$$V_{out} = -10k\Omega \left(\frac{1V}{40k\Omega} \right)$$

$$V_{out} = \underline{-0.25V} \quad \checkmark \quad 0.5$$

for. $V_{in} = [0111]$

$$V_{out} = -10k\Omega \left(\frac{0}{10k\Omega} + \frac{1V}{20k\Omega} + \frac{1V}{40k\Omega} + \frac{1V}{80k\Omega} \right)$$

$$V_{out} = -10k\Omega \left(\frac{7V}{80k\Omega} \right)$$

$$V_{out} = \underline{-0.875V} \quad \checkmark \quad 0.5$$

for. $V_{in} = [1110]$

$$V_{out} = -10k\Omega \left(\frac{1V}{10k\Omega} + \frac{1V}{20k\Omega} + \frac{1V}{40k\Omega} + \frac{0}{80k\Omega} \right)$$

$$V_{out} = -10k\Omega \left(\frac{7V}{40k\Omega} \right)$$

$$V_{out} = \underline{-1.75V} \quad \checkmark \quad 0.5$$

(a). Colpitts oscillators are oscillator whose its capacitors arranged in series (two capacitor used) with parallel to one inductor. WHILE Hartley oscillators are oscillator whose its inductors arranged in series with parallel to one capacitor.

(b). Given.

$$C_1 = 0.1 \mu F \text{ and } C_2 = 0.01 \mu F,$$

$$L = 50 \text{ mH},$$

from,

$$f_r = \frac{1}{2\pi\sqrt{LC_T}}$$

$$C_T = \frac{C_1 C_2}{C_1 + C_2} = \frac{0.1 \mu F \times 0.01 \mu F}{0.1 \mu F + 0.01 \mu F}$$

$$C_T = \frac{0.001 \mu F}{0.11}$$

$$C_T = 9.09 \times 10^{-9} \text{ F}$$

Then,

$$f_r = \frac{1}{2\pi\sqrt{50 \text{ mH} \times 9.09 \times 10^{-9} \text{ F}}}$$

$$f_r = 7465.03 \text{ Hz}$$

∴ The frequency is 7465.03 Hz.

3. (a). Conditions required for a circuit to oscillate.

(i) The voltage gain must be equal to 1.

(ii) The phase angle must be 0° .

